

LLMs for Multi-Agent System Control Project Proposal

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1 Motivation

Multi-agent systems (MAS) have vast potential for various applications of robotics. From search and rescue [1], to warehouse logistics [2] MAS can be used to improve efficiency and effectiveness in various domains. However, one challenge with MAS is the significant overhead in deployment and developing such systems.

Because deployment of MAS is often a complex and expensive task, most of the body of research in MAS is completed in simulation. Specifically, a large body of research exists in real-time strategy games (specifically StarCraft) [3]. Within this body of research, the game defines a set of constraints and actions that agents can take while providing reinforcement learning algorithms significant feedback on the performance of the agents [4]. Because of the success of training agents in games, there has become a growing interest in applying these techniques to more realistic and increasingly complex scenarios [5]. However, training agents in games is still complex and expensive and many real-world deployments require a human in the loop to ensure that the agents are performing as expected. This is especially true in scenarios where the agents are operating in a dynamic environment with changing constraints and objectives.

Branching out of the need for a human in the loop, there has been a growing interest in using large language models (LLMs) to control agents in MAS. In most research, this LLM control has taken the form of high level task allocation, mid level motion planning, and low level action generation [6]. For this project, we will be focusing on the high level task allocation problem. This problem has increasingly been shown to be accomplishable by LLMs, with LLMs being able to generate task allocation plans for MAS in a variety of standard MAS and swarm problem formulations (such as flocking and formation control) [7], [8], [9] as well as robot soccer [10]. Although many different approaches have been attempted for LLM control of MAS, a particularly interesting approach involves having LLMs generate plans using a sort of intermediate representation. In [11], the authors have the LLM generate a behavior tree for the agents to follow. In [12], the authors use Linear Temporal Logic (LTL) as an intermediate representation for the LLM to generate plans.

While LLMs have been shown to be able to generate plans for MAS, there has been little work done in evaluating these systems with a human in the loop and in situations with many robots. First, much of the previous work has focused on either pre-defined and common MAS problems (such as flocking and formation control) [9] or on a small number of robots without a human in the loop [7], [8], [13]. Second, there has been little work done in evaluating the user-side metrics of these systems. In [12], the authors conducted a 4 participant user study to evaluate the usability of their system. However, this study was limited to the performance of the system and did not evaluate the user experience of the participants.

Therefore, in this project, we will be evaluating and comparing the performance and user-experience of a traditional real-time strategy game interface and a LLM interface for controlling a MAS in a simulated game environment. Specifically, we are interested in investigating whether utilizing an LLM interface for controlling the MAS leads to decreased task load and what impact the LLM interface has on performance. Additionally, much of the previous research on using LLMs to control MAS has focused on contrived examples so we are additionally interested in investigating how humans interact and perceive the LLM interface when controlling a MAS in a more complex scenario.

2 Proposed Methodology

The main goal of our project is to create a real-time strategy game interface and an LLM interface for controlling a MAS in a game environment. We will then perform a **within-subjects** user study to evaluate the performance and user experience of the two interfaces (**counterbalanced** against the interface selection). We would specifically like to compare the performance and user experience of the two interfaces in a within-subjects design to try to control for the **confounding variable** of real-time strategy game experience so that user performance can be compared comparatively against oneself. Specifically, we will be evaluating the performance of the MAS in terms of the score achieved in the game and the user experience of the participants in terms of task load and usability. To measure task load, we will be using the **NASA Task Load Index (NASA-TLX)** [14]. To measure usability, we will be using the **System Usability Scale (SUS)** [15]. Additionally, we will be collecting qualitative feedback from the participants in the form of a **Semi-Structured Interview** to gain insight into their experience with the two interfaces. Our **Research Hypothesis** is that the LLM interface will lead to a decreased task load and no meaningful difference in performance when compared to the real-time strategy game interface.

The first major part of our project is creating a game environment for the MAS to operate in. We will be using the Robotarium [16] as our MAS platform and will be creating a game environment for the MAS to operate within. Our main goals for the game environment are to create a game that is challenging enough to require coordination of a large number of agents and that includes a both positive and negative rewards. Specifically, much of the previous work has focused on contrived examples that do not require conforming to a set of constraints, whereas these are incredibly important aspects of real-world MAS problems. Therefore, our game environment will include a set of negative feedback constraints that must be communicated to the MAS by the human operator. Additionally, we will be creating a scoring system for the game that will provide positive feedback to the MAS for completing tasks and negative feedback for failing to complete tasks or violating constraints.

With the Robotarium providing a software simulator, hardware, and runtime environment for the MAS, we will be able to focus on creating the game environment and interfaces for controlling the MAS. The team has not yet decided on the specific LLM for use in the project, but selecting the LLM will be one of the first tasks for the project. Additionally, we will be creating a traditional real-time strategy game interface for controlling the MAS. A technology stack has also not yet been selected for the real-time strategy game interface, but the design is meant to adhere to the human-computer interaction principles for real-time strategy games as presented in [17] and [18].

3 Proposed Evaluation

The project is considered successful if the following criteria are met:

- A game environment is created utilizing the Robotarium platform
- A real-time strategy game interface is created for controlling the MAS
- An LLM interface is created for controlling the MAS
- A user study is conducted to evaluate the performance and user experience of the two interfaces

4 Project Timeline

4.1 Project Check-In Day 1

By project check-in day 1, the following milestones should ideally be completed:

- The game environment should be established and built within the Robotarium platform.
- The real-time strategy game interface should be designed and partially implemented
- The LLM interface should be designed and partially implemented

4.2 Project Mid-Point "Poster" Session

By the project mid-point poster session, the following milestones should ideally be completed (in addition to the milestones above):

- The real-time strategy game interface should be fully implemented and functional
- The LLM interface should be fully implemented and functional
- The user study should be designed and ready to be conducted
- The mid-point poster should be completed and presented

4.3 Project Check-In Day 2

By project check-in day 2, the following milestones should ideally be completed (in addition to the milestones above):

- The user study should be conducted and completed

4.4 Final Project Presentations

By the Final project presentation, the following milestones should ideally be completed (in addition to all other milestones above):

- The user study should be analyzed and the results should be presented
- The final project poster should be completed and presented

5 Proposed Grading Rubric

Proposed Grading Rubric	
Finding / System Component / Capability / Validation Test	Points
Game Design and Implementation	30
Real-Time Strategy Interface	30
LLM Interface	30
User Study Setup	30
User Study Data Analysis and Interpretation	30

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